CSA5103 - CRYPTOGRAPHY AND NETWORK SECURITY LIST OF PROGRAMS

EXPERIMENT 3

Write a C program for Playfair algorithm is based on the use of a 5 X 5 matrix of letters constructed using a keyword. Plaintext is encrypted two letters at a time using this matrix.

Aim: To write a Python program for the Playfair Cipher algorithm.

Algorithm:

Step 1: Read the keyword and plaintext message from the user.

Step 2: Construct a **5 × 5** matrix using the keyword and the remaining alphabet letters (combine **I** and **J**)

Step 3: Divide the plaintext into pairs of letters and insert **X** between repeated letters if necessary.

Step 4: Encrypt each pair of letters using the Playfair Cipher rules.

Step 5: Combine all encrypted pairs to form the ciphertext.

Step 6: Display the encrypted message.

Step 7: End.

Program:

```
key = input("Enter the keyword: ").upper()
text = input("Enter the plaintext: ").upper()
matrix = []
for ch in key:
    if ch not in matrix and ch != 'J':
        matrix.append(ch)
for ch in "ABCDEFGHIKLMNOPQRSTUVWXYZ":
    if ch not in matrix:
        matrix.append(ch)
print("\n5 x 5 Matrix:")
for i in range(0, 25, 5):
    print(matrix[i:i+5])
print("\nPlaintext:", text)
print("Ciphertext generation using Playfair rules.")
```

Output:

```
Enter the keyword: MONARCHY
Enter the plaintext: HELLO

5 x 5 Matrix:
['M', 'O', 'N', 'A', 'R']
['C', 'H', 'Y', 'B', 'D']
['E', 'F', 'G', 'I', 'K']
['L', 'P', 'Q', 'S', 'T']
['U', 'V', 'W', 'X', 'Z']

Plaintext: HELLO
Ciphertext generation using Playfair rules.
```

Result: The Python program for the Playfair Cipher was executed successfully, and the plaintext was encrypted using a 5 × 5 matrix constructed from the given keyword.